Journal of Manufacturing Systems 73 (2024) 349–363

Contents lists available at ScienceDirect

Journal of Manufacturing Systems

journal homepage: www.elsevier.com/locate/jmansys

Review

Unlocking the power of industrial artificial intelligence towards Industry 5.0: Insights, pathways, and challenges

Jiewu Leng [a], Xiaofeng Zhu [a], Zhiqiang Huang [a], Xingyu Li [b], Pai Zheng [c], Xueliang Zhou [d], Dimitris Mourtzis [e], Baicun Wang [f], Qinglin Qi [g], Haidong Shao [h], Jiafu Wan [i], Xin Chen [a], Lihui Wang [j], Qiang Liu [a,*]

[a] Guangdong Provincial Key Laboratory of Computer Integrated Manufacturing, and State Key Laboratory of Precision Electronic Manufacturing Technology and Equipment, Guangdong University of Technology, Guangzhou 510006, China
[b] School of Engineering Technology, Purdue University, West Lafayette, IN 47907, USA
[c] Department of Industrial and Systems Engineering, The Hong Kong Polytechnic University, 999077, Hong Kong, China
[d] Department of Electrical and Information Engineering, HuBei University of Automotive Technology, Shiyan 442002, China
[e] Laboratory for Manufacturing Systems and Automation, Department of Mechanical Engineering and Aeronautics, University of Patras, Rio Patras 26504, Greece
[f] State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University, Hangzhou 310027, China
[g] School of Automation Science and Electrical Engineering, Beihang University, Beijing 100191, China
[h] College of Mechanical and Vehicle Engineering, Hunan University, Changsha 410082, China
[i] School of Mechanical and Automotive Engineering, South China University of Technology, Guangzhou 510641, China
[j] Department of Production Engineering, KTH Royal Institute of Technology, Stockholm, Sweden

ARTICLE INFO

Keywords:
Industrial artificial intelligence
Collaborative intelligence
Self-learning intelligence
Crowd intelligence
Industry 5.0

ABSTRACT

With the continuous development of human-centric, resilient, and sustainable manufacturing towards Industry 5.0, Artificial Intelligence (AI) has gradually unveiled new opportunities for additional functionalities, new features, and tendencies in the industrial landscape. On the other hand, the technology-driven Industry 4.0 paradigm is still in full swing. However, there exist many unreasonable designs, configurations, and implementations of Industrial Artificial Intelligence (IndAI) in practice before achieving either Industry 4.0 or Industry 5.0 vision, and a significant gap between the individualized requirement and actual implementation result still exists. To provide insights for designing appropriate models and algorithms in the upgrading process of the industry, this perspective article classifies IndAI by rating the intelligence levels and presents four principles of implementing IndAI. Three significant opportunities of IndAI, namely, collaborative intelligence, self-learning intelligence, and crowd intelligence, towards Industry 5.0 vision are identified to promote the transition from a technology-driven initiative in Industry 4.0 to the coexistence and interplay of Industry 4.0 and a value-oriented proposition in Industry 5.0. Then, pathways for implementing IndAI towards Industry 5.0 together with key empowering techniques are discussed. Social barriers, technology challenges, and future research directions of IndAI are concluded, respectively. We believe that our effort can lay a foundation for unlocking the power of IndAI in futuristic Industry 5.0 research and engineering practice.

1. Introduction

Recently, Industry 5.0 has been proposed by the European Commission to achieve a human-centered, sustainable, and resilient European industry [1]. The human-centered approach places workers' requirements, preferences, and interests at the center of the manufacturing process, moving from a technology-driven paradigm to a human-centered and society-centered one. Sustainable development requires the development of circular processes, recycling of resources, reducing waste and environmental negative impacts, and thus forming a circular economy with higher resource efficiency. On another hand, the United Nations (UN) has released a series of goals to be achieved, also known as Sustainable Development Goals (SDGs). Therefore, the above-mentioned SDGs can be considered the goals for the realization of Industry 5.0, and by extension Society 5.0. Resilience necessitates the cultivation of greater responsiveness within the manufacturing system

* Corresponding author.
E-mail address: liuqiang@gdut.edu.cn (Q. Liu).

https://doi.org/10.1016/j.jmsy.2024.02.010
Received 16 October 2023; Received in revised form 1 February 2024; Accepted 16 February 2024
Available online 29 February 2024
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to better withstand disruptions and ensure that critical infrastructure can be provided and supported in times of crisis [2].

The significance of Industry 5.0 is to match humans with intelligent machines and train collaborative robots to adapt to the intentions of humans when working together. Human intelligence can be used for critical thinking about customized demands, and IndAI systems and collaborative robots can be used for labor-intensive tasks, thereby effectively alleviating the weakening of manpower and improving production efficiency [3]. Industry 5.0 will change the global manufacturing system, infiltrating a new generation of IndAI systems into the supply chain and production shop floor, and introducing more cost-effective and human-centric IndAI.

The combination of AI and industry is proceeding. IndAI focuses on developing, validating, and deploying various intelligent algorithms for industrial applications with sustainable performance [4–9]. From the perspectives of social science, management & operations science, and computer science, the research of AI should be a hundred schools of thought, such as embodied AI, physical AI, and artificial social intelligence, which are indeed reasonable in the corresponding context. However, in industry, the simple pursuit of more advanced AI is not what we want to emphasize at the moment. In the process of digital transformation and upgrading of the industry, due to either the absence of value-oriented awareness or the lack of systematic understanding of IndAI technologies, there is often a contractionary that the design of IndAI models/algorithms does not match the application scenario. And it includes three typical improper designs/usages of IndAI models/algorithms:

1) Abuse of IndA: Using the IndAI models to solve the problems could be potentially addressed by clever design of functional materials, mechanical structures, and system behaviors. Physical hardware design usually is more reliable and cost-effective than software and codes. Abuse of IndAI models in industrial controls may lead to an increase in system complexity as well as an increase in system failure possibility.
2) Complicating simple scenarios: Deploying advanced IndAI in simple scenarios can meet most of the intelligent decision needs of the scenario. The indiscriminate pursuit of high-end intelligence not only makes computing performance surplus but also increases the implementation cost burden.
3) Simplifying complicated scenarios: Only low-end or inadequate IndAI are equipped in complex scenarios. Although the cost of IndAI implementations is reduced, this may result in wrong decisions, quality issues, and yield loss.

The first two situations resulted from a technology-driven perspective, without sufficient consideration of the value-oriented evaluation of the system sustainability and societal needs. With the continuous upgrading of the industry and the emerging penetration of Industry 5.0, more and more IndAI technologies will be applied in industrial processes, but a relatively systematic discussion of the IndAI framework for guiding the value-oriented design and implementations is rare.

The motivation of this perspective article is to address the improper design and usage of IndAI in the manufacturing paradigm evolution towards 5.0, and the goal is to present insights on the IndAI for providing guidance rather than review all IndAI papers to make a typical statistical survey. Although it is not necessary to address the detailed literature review methodology as a survey, certain guidance about the literature research is included here to relate the previous works to this article. Although many studies on intelligent manufacturing and specific models of AI such as deep learning and machine learning applications may involve research on IndAI, using these keywords will produce vast papers, that are similar in the application logic and could be categorized into well-known sub-fields from different perspectives. Therefore, to highlight the focus on unlocking the power of IndAI, two keywords, namely, "industrial artificial intelligence", and "'Industry' and 'Artificial Intelligence'", were used for retrieving publications from the Web of Science database. This inclusive search outputs 17 and 316 publications, respectively, for further screening (up to 30 Sept. 2023). Secondly, high-quality articles advancing IndAI models/methods/technologies (e.g., smart design, intelligent manufacturing, and predictive maintenance) are included. As a result, 103 publications that satisfy the selection criteria were used. Additionally, 32 supplementary publications discussing the trending of AI/IndAI (e.g., collaborative intelligence, crowd intelligence, and self-learning intelligence) and related empowering technologies for AI/IndAI (e.g., robots and CPUs) were summarized to highlight and guide the discussion focus on unlocking the power of IndAI towards Industry 5.0. Therefore, this article includes 135 references in total.

This perspective article classifies the IndAI by rating the intelligence levels, which provides a guide for proper designing and implementing appropriate models and algorithms to unlock the power of IndAI in practice. Key characteristics and the empowering technologies of IndAI for human-centric, resilient, and sustainable manufacturing towards Industry 5.0 vision are identified. Then, typical pathways of IndAI applications are discussed in five dimensions, namely, design, manufacturing, marketing and logistics, operation and maintenance service, and recycling and remanufacturing. Finally, social barriers, technical challenges, and future research directions of IndAI for Industry 5.0 are concluded.

This perspective article is outlined as follows. Section 2 presents the insights of IndAI towards Industry 5.0. Section 3 describes the pathways of IndAI towards Industry 5.0. Section 4 analyzes the challenges and research directions. Finally, Section 5 concludes.

2. Insights for unlocking the power of IndAI towards Industry 5.0

2.1. Connotations of IndAI

Since the Dartmouth Conference in 1956, AI has been studied by a large number of scholars. AI technologies are categorized by their capacity to mimic human characteristics, the technology they use to do this, their real-world applications, and the theory of mind. Since Hinton et al. put forward deep learning in 2006, AI began its prosperous development, including its applications in industry. AI is evolving from AI 1.0 to AI 2.0 [10]. The initial definition of AI 2.0 is a new generation of AI based on the significantly changed new informatization environment, a high level of autonomy and decision-making capability, and the development of new goals. The new informatization environment is driven by the popularization of the mobile Internet and advanced computing terminals, the penetration of sensor and control networks, the rise of online social communities, and the emergence of a new generation of IT (e.g., blockchain, cloud computing, and big data analytics [7]). The core concepts of AI 2.0 recognized by the community include big-data-based Intelligence, cross-media Intelligence, Internet-crowd intelligence, human-machine hybrid-augmented intelligence, and autonomous intelligence. These five intelligent technologies represent the birth of a new generation of AI, followed by other emerging concepts such as embodied AI, physical AI, and artificial social intelligence. The implementation of a new generation of AI in industrial processes accelerates the development of products.

The combination of AI and industry is proceeding. Some scholars presented definitions of IndAI from different perspectives as shown in Table 1. For instance, Lee et al. (2018) [4] defined IndAI as a systematic discipline, that focuses on developing, validating, and deploying various machine learning algorithms for industrial applications with sustainable performance. Based on the review of the literature and the understanding of actual industrial applications, this perspective article defines IndAI as follows: *IndAI is an interdisciplinary research field that uses advanced computer technology to deeply integrate industrial infrastructures with advanced AI models/methods and enables industrial systems to have*

Table 1
Existing definitions of IndAI.

Authors	Definitions
Lee et al. (2018)[4]	Focus on developing, validating, and deploying various machine learning algorithms for industrial applications with sustainable performance.
Zhang et al. (2019) [11]	A deep integration of AI technology and industrial process, in which industrial enterprises take IndAI technologies to achieve intelligent function in all stages of the industrial value chain.
Peres et al. (2020) [12]	A systematic discipline focusing on the development, validation, deployment, and maintenance of AI solutions (in their varied forms) for industrial applications with sustainable performance.
Hoffmann et al. (2020)[13]	The combination of sensors with the Industrial Internet of Things, (big) data, algorithms, and processing power.
Yang et al. (2021) [14]	To combine general AI technology with specific industrial scenarios in order to achieve innovative applications, such as design model innovation, intelligent production decision-making, and optimal resource allocation.
Mathew et al. (2023) [15]	A systematic strategy for creating, validating, and deploying various machine learning algorithms for high-performing industries.
Viola et al. (2023) [16]	Any application of AI related to the physical operations or systems of an enterprise.
Rožanec et al. (2023) [17]	A specific case of Narrow AI applied to industry.

enhanced capabilities of perception, cognition, learning, organizing, execution, self-decision-making, and mutual-adaptation, and makes it empower the product/service lifecycle management to help enterprises realize the value-adding.

2.2. Classification of IndAI and its four implementation principles

IndAI is applied to different stages of product/service lifecycle activities and various levels of industrial systems, resulting in different levels of intelligence [6,18–20]. Generally, there are three grades of artificial intelligence, namely, Artificial Narrow Intelligence (ANI), Artificial General Intelligence (AGI), and Artificial Super Intelligence (ASI). Morris et al. [20] (Google DeepMind research team) defined five levels of AGI, including Emerging AGI, Competent AGI, Expert AGI, Virtuoso AGI, and Superhuman AGI. Rajani classifies AI into four categories [19]: 1) optimal in which it is not possible to perform better, 2) strong super-human which performs better than all humans, 3) super-human which performs better than most humans, and 4) sub-human which performs worse than most human. These classification criteria are designed based on the comparison between computer and human intelligence. Therefore, it is challenging to distinguish an AI model/algorithm to which grade it belongs since the criteria are neither specific enough nor executable in practice.

Another perspective to classify the AI models/algorithms is to refer to the Von Neumann architecture. For instance, Liu et al. [18] proposed four criteria based on Von Neumann architecture for defining a standard intelligence system. By referring to these four criteria, to provide a guide for designing and implementing appropriate models and algorithms to unlock the power of IndAI in industrial practice, five criteria to classify the IndAI applications are identified as follows:

Criterion 1: Whether it can interact with context, that is, whether there is an input-output system.

Criterion 2: Whether the industrial system has a knowledge database that stores and updates decision rules/patterns.

Criterion 3: Whether the industrial system can constantly and automatically learn the decision knowledge from the interaction context and make self-updating of knowledge.

Criterion 4: Whether the industrial system can self-percept, self-learn, self-execute, self-organize, and self-adapt to new environments.

Criterion 5: Whether the industrial system can share self-updated knowledge and up-to-date intelligence with other heterogeneous intelligent systems.

According to these criteria, intelligence is divided into five levels from lower level to upper level, namely, reactive IndAI, Knowledgeable IndAI, Learnable IndAI, Autonomous IndAI, and Sharable IndAI, as shown in Table 2. They represent different levels/grades of intelligence in the industry, and enterprises can choose different levels of intelligence according to their actual needs.

Designing appropriate models and algorithms in the digital upgrading process of the industry is critical for unlocking the power of IndAI while ensuring cost-effectiveness and reliability. Usually, the lower-level IndAI algorithms are more reliable, cost-effective, and robust than the higher-level IndAI algorithms. Four principles of implementing IndAI algorithms could be identified as follows:

1) A designer should avoid abuse of IndAI if the needs could be resolved by innovative, clever, and dedicated mechanism design in its material, structure, and behavior dimension since the physical mechanism is more reliable than the software and code;
2) A designer should consider all possibilities and scenarios in practice, and design an IndAI model or algorithm possessing sufficient intelligence and robustness to address the challenges;
3) A designer should prioritize the adoption of a lower-level IndAI algorithm rather than a higher-level one to save cost and improve reliability if the lower-level IndAI algorithm could adequately achieve the management and control goals;
4) A designer should make the IndAI implementations more sustainable, resilient, and human-centric under the premise of satisfying the functional and performance needs.

2.3. Opportunities of IndAI for Industry 5.0

Industry 5.0 [1] places the well-being of the workers at the center of the manufacturing process by making production respect the boundaries of our planet and the harmonious symbiosis between humans and machines, to achieve societal goals beyond jobs and economic growth, and further achieve sustainable development goals of super-smart society and ecological values, which will become a robust and resilient provider of prosperity in an Industrial Community of Shared Future [21].

Three key propositions are stated in the Industry 5.0 whitepaper [1]. The first proposition is human-centricity. Industry 4.0 focuses on the feasibility and implementation of technology but pays less attention to the needs of humans. As the actual operator of the process, workers fail to play an effective role in supervision and decision-making. Therefore, in the Industry 5.0 vision, workers return to the center of the process and act as the actual decision-makers of the process. The second proposition is sustainability. Technology-driven Industry 4.0 aggravates some common social and environmental problems. Industry 5.0 emphasizes the concept of sustainable development, reusing/recycling natural resources, and reducing waste and environmental impact, so that the industrial ecology forms a good development pattern of symbiosis [22]. The third proposition is resilience. Brittleness is a system problem that is not easy to appear. Once it occurs, it will cause serious damage to production. The pandemic, for example, highlights the fragility of the current rigid hierarchy of globalized collaboration mode of production.

Table 2
Intelligence level of IndAI.

Level	*Criterion* 1	*Criterion* 2	*Criterion* 3	*Criterion* 4	*Criterion* 5
Sharable IndAI	√	√	√	√	√
Autonomous IndAI	√	√	√	√	
Learnable IndAI	√	√	√		
Knowledgeable IndAI	√	√			
Reactive IndAI	√				

How to make industrial production better resist disruptions is an important research direction of Industry 5.0. The development of a sufficiently flexible strategic value chain, modulized manufacturing system, adaptable production capacity, predictive controls, and flexible business processes are also key issues to be faced in forming the resilience of Industry 5.0.

To achieve societal goals beyond jobs and growth, and to become a resilient provider of prosperity, it is urgent to make innovations in models/methods as well as reorganize the advanced technologies/techniques in a value-oriented manner. As shown in Fig. 1, corresponding to three propositions (i.e., human-centricity, sustainability, and resilience) mentioned above, three opportunities of IndAI towards Industry 5.0, namely, collaborative intelligence, self-learning intelligence, and collective intelligence, could be identified for unlocking the power of IndAI.

2.3.1. Collaborative intelligence

Collaborative Intelligence [23,24] is defined as the joining forces of Humans and AI. Its representative facilities are various smart devices, such as collaborative robots and collaborative agents, designed to assist human activities. These smart devices can notice, understand, and feel the workers, as well as the goals and expectations that are being performed. Its purpose is to observe and learn how humans perform tasks and to help workers perform tasks. In Industry 5.0 vision, AI is required to permeate humans to collaborate, recognize, and actively enhance each other's complementary strengths: 1) the intuition, leadership, teamwork, creativity, and social skills of workers, and 2) the speed, precision, and persistence of machines [21]. Through this complementarity, the full potential of machines and humans can be fully brought into play, and human-centered collaborative intelligence can be truly realized.

2.3.2. Self-learning intelligence

Self-learning intelligence enables machines to consistently learn and create more knowledge and decision patterns by themselves [25]. The societal needs and the manufacturing system itself are dynamically changing over time, and the operating data thereby may be of new unknown patterns and are difficult to label. It requires machines to self-learn without labeled data when transferred to a new environment. Self-learning AI is artificial intelligence that can train itself using unlabeled data. On a high level, it works by analyzing a dataset and looking for patterns that it can draw conclusions from. The European Research Centre is supporting a self-learning AI research project, named GENMOD (https://erc.europa.eu).

As illustrated in Fig. 2, self-learning IndAI models can make the machines more intelligent and give them the potential to generate their own new intelligent decision patterns. These characteristics of self-learning intelligence give machines as well as the manufacturing system the ability to develop sustainability. This intelligence could change the way AI is created and change our inertial thinking to help machines/systems solve many kinds of problems that they didn't counter before. Because of the low cost of trial and error, self-learning intelligence can verify various cases to find the optimal solution for the new environment and complete the self-evolution.

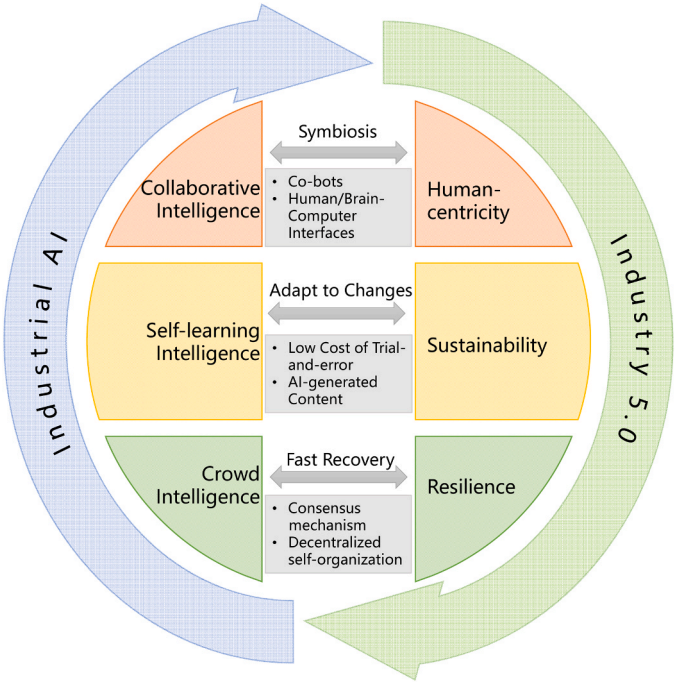


Fig. 1. Characteristics of IndAI for Industry 5.0.

2.3.3. Crowd intelligence

Crowd Intelligence [26,27] emerges from the collective intelligent efforts of massive numbers of autonomous individuals, who are motivated to carry out challenging computational tasks under a certain Internet-based organizational structure. In the context of Industry 5.0, crowd intelligence enhances the system resilience under disruptions, based on strong self-adaptivity among human-machine hybrid groups and lower bio-machine hybrid groups. Usually, researchers refer to the former intelligence as collective intelligence and the latter intelligence as swarm intelligence. Crowd intelligence is a synthesis of these two kinds of intelligence. This development of such complexes requires groups with different knowledge and complementary skills to be coordinated in a distributed manner to produce decentralized and resilient crowd intelligence. It is necessary to connect individuals into a group through decentralized collaboration, competition, and good incentive mechanisms, and at the same time, quality control of the crowd is necessary to ensure the quality of the whole system.

Fig. 3 illustrates the rationale of crowd intelligence. Individuals make use of the obtained information, produce group interaction through competition and cooperation with other individuals, and then make their own decisions. Crowd intelligence enables rapid reorganization and adaptation in response to unexpected disruptions to form a new balanced manufacturing organization [28]. Finally, the individual decisions are aggregated and fed back to the group to produce a more excellent problem solution for handling disruptions.

Despite humans, machines in the crowd could be empowered by modeling as agents [29], and its high-level potential benefits include:

1) Collaborative and effective decision system. Democratize access to IndAI models and their benefits by enabling coordinated IndAI and multi-agent interactions [8], decentralizing data ownership, sharing knowledge, and access to huge datasets;
2) Self-organizing cooperation. Enabling a new form of self-organizing intelligence beneficial to the race with networks that consider interoperability, data sovereignty, privacy, modularity, and scalability;
3) Democratic governance. Incorporating democratic governance and other incentives in ways that support behaviors that are beneficial to the network as a whole moves IndAI decentralized and free markets in a favorable direction;
4) Beneficial IndAI. On the decentralized network, beneficial reserves and network wealth form a virtuous circle through benefit ratings, so that IndAI can attract more users while exerting its value and cultivating a more beneficial user economy.

Fig. 3 shows the decentralized agent model where nodes are connected in a distributed manner and each node is characterized by a high degree of autonomy. An agent can achieve decentralized access to IndAI based on this model. When some nodes cannot be accessed due to

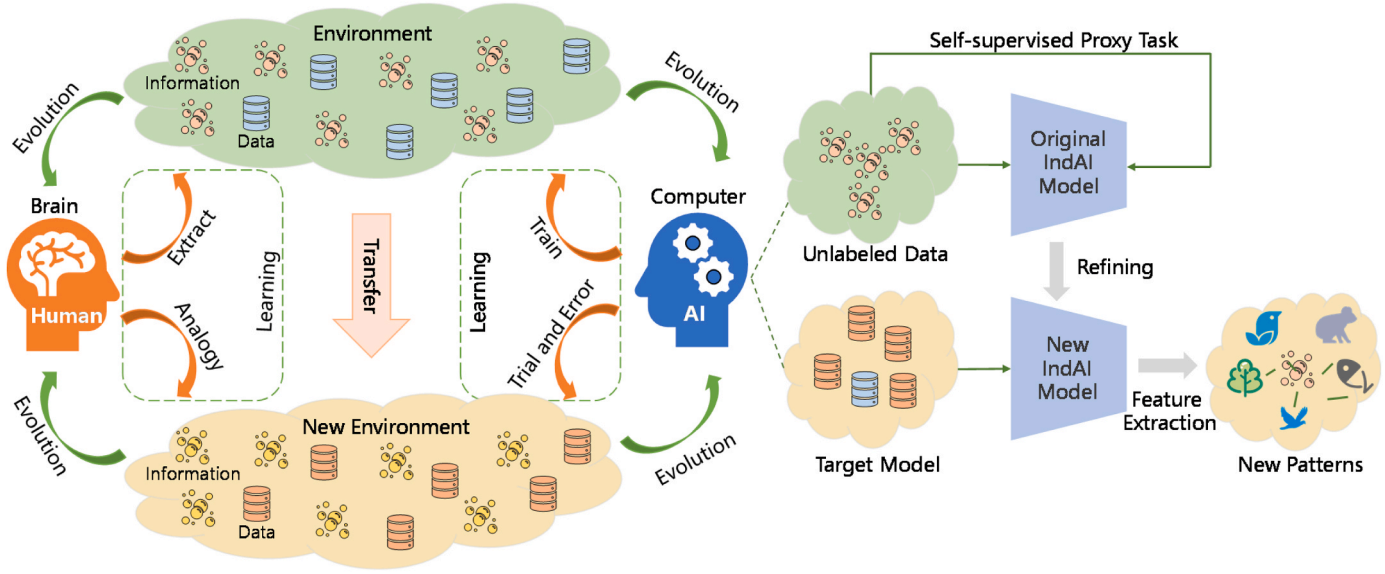


Fig. 2. Self-learning intelligence.

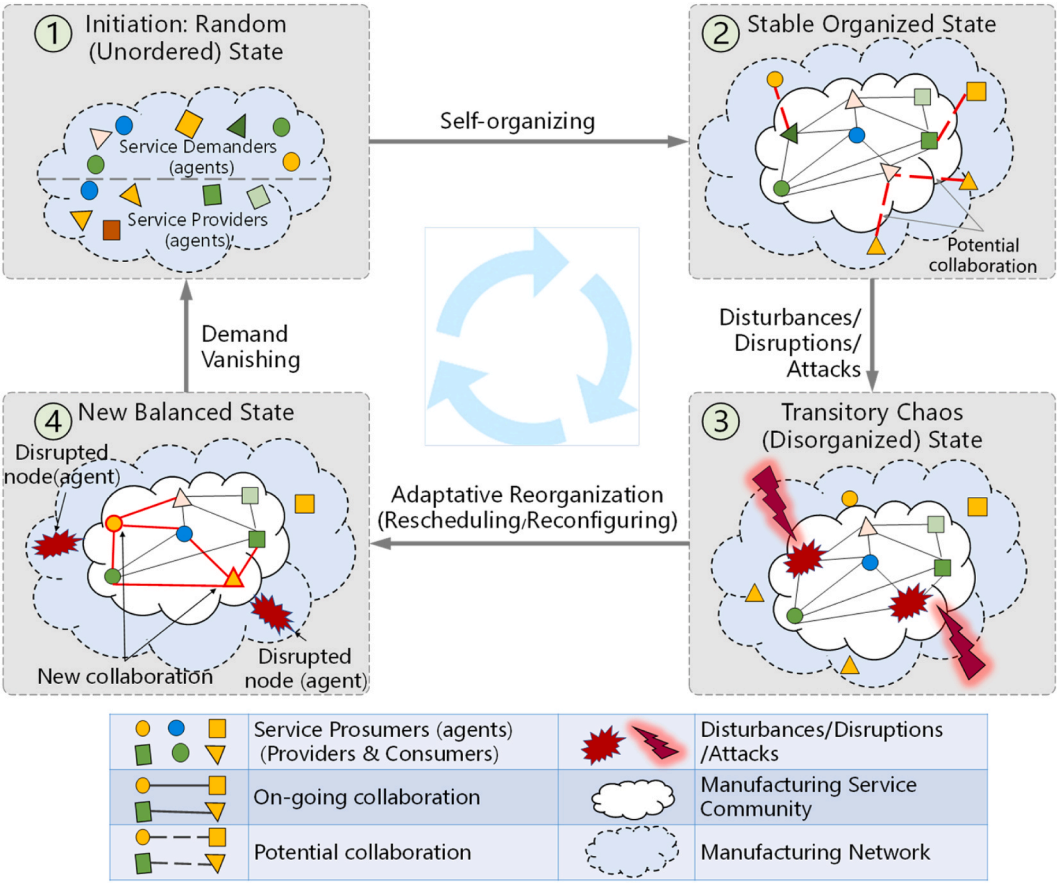


Fig. 3. Crowd intelligence for resilient manufacturing under disruptions [28].

disruptions, the missing part is compensated by the call of other nodes as much as possible to achieve resilient decentralized crowd intelligence.

Three opportunities of IndAI discussed above are promising research directions for advancing Industry 5.0 engineering practice, but they should not be directly identified as determinate directions/stages of Industry 5.0 development. Also, the simple pursuit of more advanced AI is not what we want to emphasize at the moment. Designing appropriate models and algorithms in the upgrading process of the industry is of greater significance for unlocking the power of IndAI.

3. Pathways for unlocking the power of IndAI towards Industry 5.0

3.1. Framework of IndAI applications

The development of IndAI cannot be separated from the advancement of mainstream AI, and enterprises also need a reference to guide their IndAI transformation. Therefore, this perspective article summarizes the aspects that enterprises will be involved in the IndAI applications and puts forward a technical IndAI reference framework as shown in Fig. 4. It is divided into four layers, namely, the Hardware Infrastructure Layer, Computing Engine Layer, AI Algorithms Layer, and Industrial Application Layer, together with related empowering technologies across layers.

The first layer provides the Internet of Things (IoT) facilities, computing hardware, data services, and networking capabilities. The IoT facilities are essential to quickly understanding the production status of the industry system. Networking facilities enable rapid data transmission and informative decisions.

The second layer provides the runtime environment and computing platform. A big data platform is a set of infrastructures that deals with storage and real-time computing of massive uninterrupted data. At present, there are mature methods to deal with multi-source heterogeneous data. The running environment is a container that can support the development and training of InAI models/algorithms.

The third layer includes typical core AI algorithms used in industry [5], such as linear regression (e.g., for material usage optimization), logistic regression (e.g., for simplifying tomography system), support vector machine (SVM) (e.g., for fault diagnosis [30]), Naive Bayes (e.g., for intrusion detection [31]), decision tree (e.g., for quality disturbance detection [32]), K-nearest neighbor (e.g., for traffic flow prediction [33]), K-means (e.g., for wind power prediction [34]), random forest (e.g., for machinery fault diagnosis [35]), dimensionality reduction (e.g., for wind turbine state monitoring [36]), and neural network (e.g., for flexible joint manipulator control [37]). These algorithms can be applied to various industrial scenarios.

The most popular three academic schools of core AI algorithms are Symbolicism, Connectionism, and Actionism, respectively. Each possesses distinct characteristics that align with different requirements in industrial scenarios.

1) Symbolicism AI: Symbolism is also called logicism, and its core is mathematical logic, which aims to build AI systems with axioms and logic systems. The representatives include Support Vector Machine (SVM) and Expert System. This type of AI model has achieved more results in the early stage of AI development.
2) Connectionism AI: Connectionism originated from bionics, so it is also called bionicsism. Based on the model of the human brain, it proposes to use algorithms to simulate the neurons of the human brain and connect the neurons as nodes to form neural networks. Nowadays, the neural network has been applied in various industries and has achieved good results. The development of neural networks has also developed from the initial Perceptron, Long Short-Term Memory (LSTM), Hopfield Neural Network, and Backpropagation Neural Network to the Current Deep Neural Network, Recurrent Neural Network, and Convolutional Neural Network.
3) Actionism AI: The core idea of actionism is cybernetics, which links the working principles of the nervous system with information

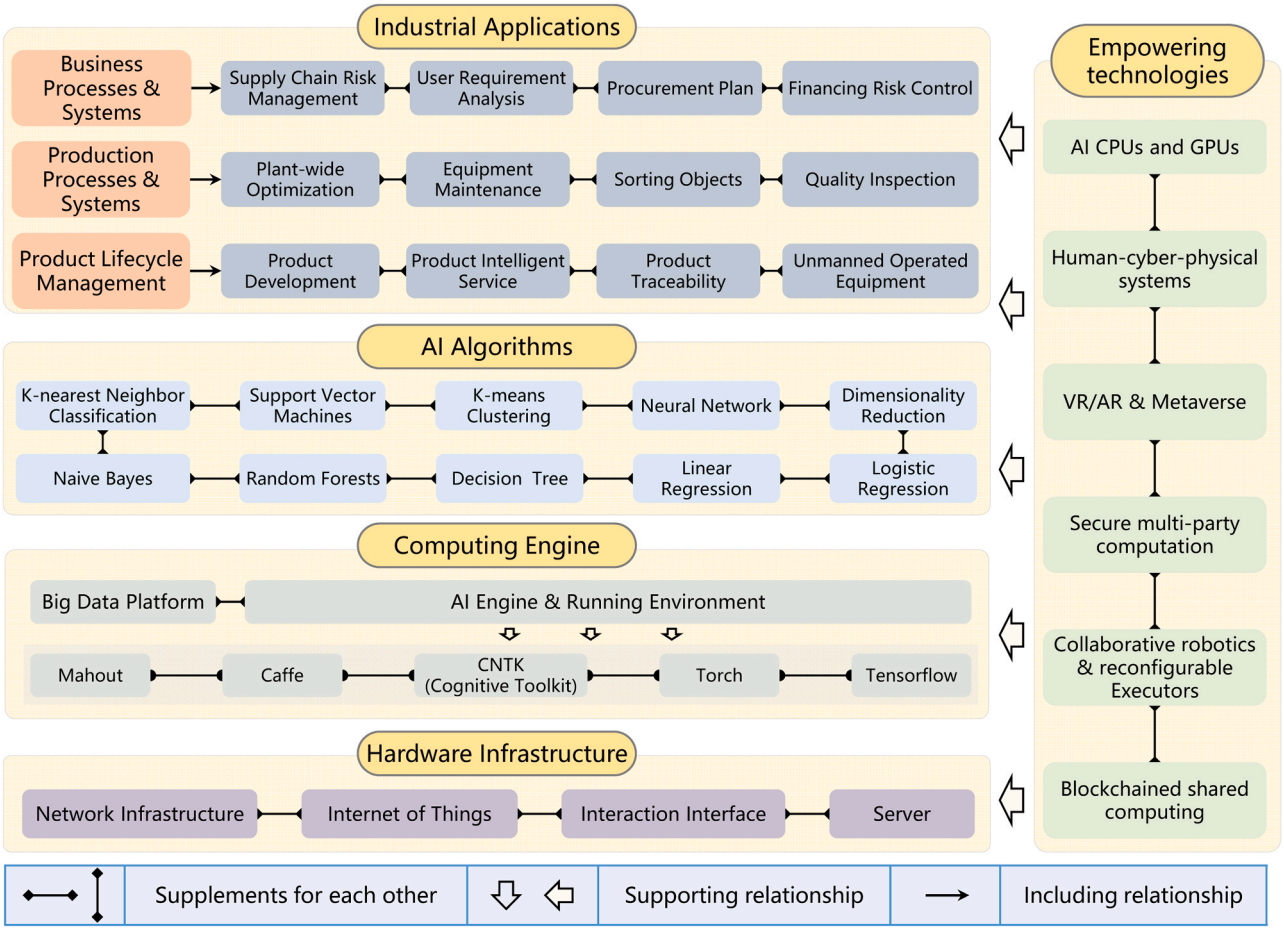


Fig. 4. Reference framework of IndAI applications.

theory, control theory, logic, and computers. A representative example of this type of AI (e.g., Reinforcement Learning, Genetic Algorithm, Evolutionary Algorithm) is the hexapod robot with a control system that simulates insect behavior based on sensory-action patterns. Actionism focuses on the development of applied technology, which is purposeful and can be quickly put into a practical application.

The industrial application layer adapts to the individualized design, configuration, and operation of production/business processes/systems, as well as product lifecycle management needs.

The reference framework of IndAI presented here is not new knowledge but a gathering of best practices. In the implementation of IndAI, the practitioners are suggested to refer to the presented framework based on four principles proposed in Section 2.2, with a systematic understanding of goals, constraints, and characteristics of their industries, to obtain a balance of the performance, cost, reliability, and security of the applications. And not all elements in the presented reference framework are supposed to be included and realized in a specific implementation, since the simpler the design, the higher the reliability of the system.

3.2. Dimensions for unlocking the power of IndAI towards Industry 5.0

Typical dimensions for unlocking the power of IndAI towards Industry 5.0 are discussed from the product lifecycle management perspective. The product lifecycle dimension comprises a series of interrelated value-creating activities from product concepting to product recycling, such as design, manufacturing, logistics & marketing, and operation & maintenance. As shown in Fig. 5, IndAI in the product lifecycle dimension accelerates the upgrading speed, production efficiency, and service quality of products. The lifecycle activities could be iteratively optimized to achieve sustainability.

3.2.1. IndAI-driven smart design

Design is a process of selecting the appropriate variables and parameters to satisfy the individualized needs of users under given constraints and existing conditions. Smart design is a process of optimizing all feasible schemes from the dimensions of function, structure, behavior, control, and performance of products/services/systems according to the requirements [38]. Design optimization models could be established for selecting mechanical system components with mixed design variables and constraints [39]. IndAI could discover the risks and uncertainties in the design optimization process [40]. Smart design involves both products and the corresponding manufacturing systems. The smart manufacturing system design is much more complex and involves multiple dimensions including function, structure, behavior, control, intelligence, and performance [41].

The smart design could reduce the amount of computation and shorten the execution time. Nowadays, AI-Generated Content (AIGC) technology is a promising enabler for empowering smart design. The generative design enabled by the IndAI is a promising direction. However, the time-consuming process of complex modeling and configuring the design problem is an obstacle hindering unlocking the power of AIGC.

3.2.2. IndAI-driven smart manufacturing

Manufacturing is a process of processing, assembling, and inspecting materials to create products. Smart manufacturing applications involve all system levels from equipment (e.g. [42]), manufacturing cell (e.g. [43]), production line (e.g. [44]), workshop (e.g. [45], [46]), factory (e.g. [47]), to supply chain management (e.g. [48]). A smart manufacturing system is a reconfigurable, flexible, agile, and resilient system with self-sensing, self-learning, and self-adaptive functions in multiple dimensions, such as process optimization, production scheduling [9], quality control, and operations management.

Nowadays, production scheduling has become an indispensable part of smart manufacturing [49], and the application of intelligent algorithms is also a reflection of IndAI. Production scheduling organizes the timing and resources of jobs or tasks in manufacturing or service to achieve predetermined system goals, such as arranging the online sequence of various products in the assembly line to achieve the shortest makespan. Scheduling intelligence is mainly reflected in various intelligent optimization algorithms and models in job shops [50]. Current intelligent scheduling studies emphasize more on energy consumption [51]. Dynamic events including machine failure, travel delay operation, and poor-quality operation make intelligent scheduling more difficult [52]. Moreover, trusted collaborative manufacturing remains challenging, and how to implement a secure and sharable IndAI model in a distributed manufacturing context is also challenging.

Enhancing the machine vision by integrating IndAI for defect recognition and fault detection is fruitful [53]. Machine learning-based

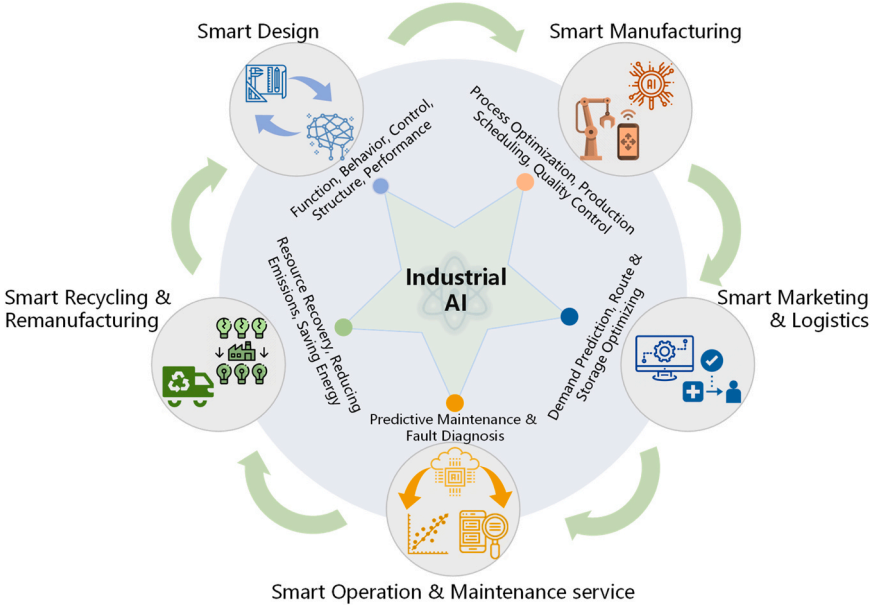


Fig. 5. IndAI empowering the product lifecycle management.

online fault identification methods could be designed for the timely identification of material defects caused by process non-conformities [54]. The application of deep neural networks to machine vision tasks enables substantial leaps in model performance [55]. Good classification can be achieved without a large number of labeled datasets. Therefore, few-shot learning-based, learnable, and physics-informed IndAI are hot topics in smart manufacturing for unlocking their power in the industry.

3.2.3. IndAI-driven smart marketing and logistics

The marketing process requires analysis, prediction, and accurate recommendations to different consumers. Smart marketing combines advanced marketing thinking with new IndAI methods. It is supposed to be a consumer-centric solution towards Industry 5.0 to achieve personalized marketing. Intelligent demand forecasting systems could be developed based on the analysis and interpretation of historical data by integrating time series analysis techniques. The periodic trend of user behavior could be mined to recommend the potential goods to the user at the appropriate time. For instance, Li et al. [56] proposed an intelligent attention allocation link prediction algorithm to predict the possible friendship between users in different circles for solving the problem of cross-marketing network sparsity. Also, smart logistics involves route optimization and storage optimization in the physical flow process of materials/products from supplying to receiving to meet the increasing complexity and volume of logistics operations due to the collaboration and integration of online and offline channels globally [57]. How to make marketing more personalized is the problem smart marketing and logistics have to face.

3.2.4. IndAI-driven smart operation and maintenance service

Operation and maintenance services encompass a series of value-adding activities resulting from the interaction between product providers and customers. The smart operation and maintenance service applications majorly include product predictive maintenance and fault diagnosis.

Predictive maintenance is to predict the future development trend of the product status and formulate corresponding maintenance plans in advance according to its operating trend and possible failure modes. IndAI model-based predictive maintenance solution could be developed to analyze the health status of products and machines [58]. For instance, Wang et al. [59] proposed a long- and short-term memory recurrent neural network-based maintenance predictor to improve the maintenance efficiency of high-speed railway power equipment. Degradation estimation methods could be also designed for redundant system maintenance planning optimization [60].

Fault diagnosis is to comprehensively understand the device condition, discover faults in time, and identify the source of faults. Integrating various machine learning strategies (e.g., transfer learning [61], deep convolutional network [62], and two-stage learning [63]) with multi-domain feature extraction methods could improve fault diagnosis accuracy.

Nowadays, smart operation and maintenance services have become more flexible, networked, autonomous, and adaptive [64]. Integrating the new generation of behavior operation management theory into hybrid IndAI models is a promising solution for unlocking their power in the industry.

3.2.5. IndAI-driven smart recycling and remanufacturing

Recycling and remanufacturing have an important impact on the circular economy to provide all-round sustainable benefits [65]. In a circular economy, the goal is to maintain utility and value throughout the life cycle of products, components, and materials. Recycling and remanufacturing are mainly reflected in several aspects, such as reusable resource recovery, reducing emissions, and saving energy. Combining consumers' low-carbon preference and remanufacturing preference with market demand in the recycling and remanufacturing decision is important for maximizing both manufacturers' and consumers' interests [66], in which IndAI could act as a key enabler. For instance, a joint optimization could make the pricing, quality level, and effort decisions of manufacturers, retailers, and third parties more reasonable and coherent [67]. The IndAI models should be integrated with innovations in the process flow, flexible additive manufacturing technologies [68], and recyclable materials usage [69]. IndAI enables new process flow for the production to achieve environmental friendliness and low energy consumption [70].

Manufacturers are committed to developing sustainable production and consumption strategies to reduce their negative impact on the environment and society [71]. Emerging IndAI technologies and advanced manufacturing capabilities are supporting the sustainable development and intelligent remanufacturing of manufacturers [72].

Table 3 summarizes the specific IndAI applications in product lifecycle management for unlocking their power in the industry. In each implementation dimension of IndAI, the practitioners are suggested to refer to the presented framework based on four principles proposed in Section 2.2, with a systematic understanding of goals, constraints, and characteristics of their industries, to obtain a balance of the performance, cost, reliability, and security of the applications.

3.3. Related empowering technologies for implementing IndAI

The combination of IndAI models/algorithms and related empowering technologies can produce outstanding results for unlocking their power in the industry.

3.3.1. AI CPUs and GPUs

AI CPUs and GPUs are of dedicated design for fasting IndAI computing. CPU is mainly used to manage the entire system, execute software programs, process input and output, and control task scheduling in AI [76]. It is responsible for the operation of the AI framework, data preprocessing, and post-processing tasks. CPU is very important for controlling processes and completing complex logical calculations, especially in model optimization, parameter updating, and adjustment during training. On another hand, GPU is mainly used for large-scale parallel computing in AI, especially good at dealing with large-scale data computing tasks such as matrix operations and deep neural

Table 3
IndAI applications for product lifecycle management.

Lifecycle	Cases	Ref.
Smart Design	Constrained design optimization of function, structure, behavior, control, intelligence, and performance of products/services/systems Fuzzy multi-objective evaluation of design decision, Design optimization for manufacturing and assembly.	[39],[73], [40]
Smart Manufacturing	Multi-objective flexible job shop scheduling, Online fault identification of manufacturing system operating, Product surface defect detection.	[74],[75], [51],[54], [55]
Smart Marketing and Logistics	Intelligent demand forecasting, Mining of the periodic trend of user behaviors, Intelligent attention allocation link prediction and path optimization, Warehousing optimization in the logistics process.	[56],[57]
Smart Operation and Maintenance	Predictive maintenance, Machine fault diagnosis, Proactive adjustment of operating parameters.	[59],[58], [62],[61]
Smart Recycling and Remanufacturing	Repairing damaged parts, Fully recycling waste, Fast remanufacturing decision.	[68],[70]

networks. Because neural networks play a key role in AI, the parallel computing power of GPU can accelerate the training and reasoning process of neural networks. GPU's high-performance parallel computing can process multiple data points at the same time, providing faster training and reasoning speed, thus accelerating the response time of AI applications. In general, CPU is suitable for general AI computing tasks, while the GPU is suitable for graphics and large-scale parallel AI computing tasks [77]. This collaboration enables AI applications to run and process large amounts of data more efficiently, improving overall computational performance and efficiency.

The development of CPU/GPU hardware is the basis of AI deep applications [78]. Due to the influence of structure and other factors, CPU will produce a lot of heat during operation, which will affect the operation efficiency of AI [79]. Therefore, the thermal management system of CPU is challenging [80]. To improve its operating efficiency, for example, Arafa et al. [81] proposed an extensible and accurate simulation framework that enables GPU code developers and architects to accurately predict the performance of applications on different GPU architectures. To make full use of GPU's powerful 2D/3D hardware-accelerated graphics rendering performance or parallel computing potential, Tan et al. [82] proposed a multi-channel GPU virtualization architecture, established the corresponding credit allocation and transfer mechanism model, and redesigned the virtual multi-channel GPU fair scheduling algorithm. To make full use of the available heterogeneous hardware, Chrysogelos et al. [83] introduced hetexchange, a parallel query execution framework, which encapsulates the heterogeneous parallelism of modern multi-CPU-multi-GPU servers and supports the parallelism of existing sequential relational operators.

In general, the rational use, architecting, and coordinating of CPU and GPU is challenging to improve the performance and response speed of IndAI for specific industrial applications.

3.3.2. Human-cyber-physical systems

The blueprint of Industry 5.0 places humans as the center of the manufacturing system, in which the human-cyber-physical system (HCPS) and cyber-physical-social system (CPSS) play important roles. From the interaction aspect, HCPS enhances workers' abilities to interact with machines via intelligent human-machine interfaces and techniques designed to adapt to the workers' cognitive and physical requirements. From the cognition aspect, HCPS enhances workers' physical-, sensing- and cognitive- capabilities via wearable devices [84]. Operators in HCPS could be integrated into flexible manufacturing processes and outlined what kind of HCPS-based infrastructure should be designed to enhance workers' efficiency in production systems [85]. The next generation of HCPS for intelligent manufacturing is supposed to make the role of humans more prominent and to form the human-machine symbiosis form [86].

As an enabler of HCPS, digital twin simulates, verifies, predicts, and controls the lifecycle process of physical systems based on real-time data analytics, and finally realizes an optimal closed-loop for intelligent decision-making. Digital Twin provides runtime and real-time data support to the IndAI models. In the industry, digital twin-empowered IndAI applications usually focus on improving system design, configuration, and operation optimization. In the system design stage, digital twins could empower IndAI for the individualized design of the structure, behavior, control, and performance model of smart manufacturing systems [87]. In the system configuration stage, digital twins could empower IndAI for rapid semi-physical commissioning and controlling of smart manufacturing systems [88,89]. In the system operation stage, digital twins could empower IndAI for equipment health management [90], as well as remaining useful life prediction [91]. By empowering digital twins with intelligence, the status of the equipment can be summarized to provide a basis for the rational arrangement of operation and maintenance.

CPSS emphasizes the social dimension [92]. In the case of more open manufacturing models and growing demand for individualized manufacturing, social manufacturing is a new industrial paradigm. Through CPSS, various communities can form complex dynamic autonomous systems to jointly create individualized products and services. CPSS could empower IndAI to realize group-centered intelligent recommendations [93]. CPSS could be also designed to empower IndAI to provide high-quality active and personalized services for humans [94].

Both HCPS and CPSS provide services around people, reflecting the human-oriented idea. To some extent, HCPS and CPSS are mutually explicable. HCPS takes humans as a bridge and cooperates with humans through IndAI to facilitate symbiotic deep human-machine collaborative work for intelligent manufacturing. CPSS extends this focus on people to the social level, incorporates social organizations into the system, and enables resources to be deeply developed and utilized. In the integration of HCPS/CPPS with IndAI, there are still some potential challenges on the security and privacy concerns to be solved before unlocking the power of IndAI.

3.3.3. Metaverse

Industry 5.0 attaches importance to the development of the workforce and will create more jobs in the areas of intelligent systems, AI, human-robot collaboration, behavior operation analysis, predictive maintenance, and the design/training/scheduling/reuse of cobots/robots [95]. New skills emerge, including robot assistant, machine programming, robot managing, cyber-physical systems analysis, collaborative operating, and virtual operating. This means that the importance of worker education and training systems will be raised to a new level [96], in which augmented reality and metaverse technologies could be used to provide risk-free training for the workforce, which accelerates the growth of workers.

Augmented Reality (AR) combines and interacts with the virtual space with the real-world scene through actuarial calculation and image analysis. AR is currently mainly used in product assembly and maintenance. For the maintenance of relatively concealed parts such as aircraft wheels [97], it is impossible to photograph clear damaged parts for professional engineers to identify. The AR remote maintenance can be quickly identified and specific maintenance work can be started. Compared with the traditional method, it reduces maintenance time, reduces transportation costs, and improves maintenance efficiency. Projector-based spatial AR systems in quality assurance could improve the accuracy of positioning in machining [98]. Cloud computing-based AR could empower IndAI for predictive maintenance [99]. Even as AR matures, it still faces some challenges. The ergonomic design of the AR system should enable users to achieve a high level of satisfaction and avoid long-term visual and neck fatigue. Since AR equipment is usually in direct contact with the human body during use, its safety must be improved during human-machine collaboration.

The metaverse is regarded as the ultimate form of the Internet. The virtual world coexists with the real world. Virtual users can obtain the same interactive experience as in reality, and the virtual world also has its economic systems. Virtual users can also conduct economic management in the virtual world. Metaverse aligns with the vision of Web 4.0, representing a digital ecosystem where individuals and organizations collaborate in a human-centric approach to create personalized value [100,101]. At present, the metaverse has great application prospects in biology [102] and tourism [103]. Metaverse may be simply defined as Persistent Virtual Reality in a narrow sense, and the advanced stage and long-term vision of Digital Transformation in a broad sense. The excellent performance of the metaverse in virtual space accelerates the personalized value creation of the manufacturing industry and makes Industry 5.0 possible [104]. Their standards, architecture patterns, communication protocols, data processing, reliability, and integration with IndAI models will be the focus of subsequent research when unlocking their power.

3.3.4. Secure multi-party computation

Secure Multi-Party Computation (SMPC) uses the Internet as the carrier to realize the on-demand acquisition of secure computing resources for distributed data. SMPC in the industry is usually architected in a cloud-edge orchestration manner [105]. The cloud-side computing performs the costly large-scale IndAI model training, while edge-side computing executes the simple reasoning in IndAI algorithms to improve efficiency [106], resource utilization[107], and accuracy [108]. Intelligent resource allocation schemes in SMPC are critical to adaptively allocate computing and networking resources for reducing the average service time [109]. SMPC not only improves decision quality via the cloud but also speeds up decisions on the edge.

The security and privacy of computing in the cloud is an urgent problem to be solved. Blockchain is essentially a decentralized secure computing paradigm that could be used to enable reliable SMPC to realize product lifecycle management. The tamper-proofing ability of blockchain is usually used to establish an authenticity verification network for the distributed production, circulation, and consumption of products to improve credibility and trustiness [110,111]. Scholars used blockchain to reshape the industrial IoT to improve the security and privacy of data in the information interaction in collaborative design and manufacturing to achieve trusted cooperation between the manufacturer, the designer, and the customers [112]. The traceability of spare parts and product quality in industries is much more complex than the traceability of goods in the supply chain. Industrial blockchain application is still in the preliminary exploration stage in the traceability of industrial product quality. Integration self-organizing multi-agent system with blockchain at the edge of SMPC enables autonomous decision-making and distributed cooperation among agents with high efficiency [113]. Data in blockchained SMPC applications are usually of high quality and thereby could improve the accuracy of IndAI models in the cloud since it has been verified by multiple participants. However, there are still many potential problems, such as transmission delay caused by the consensus algorithms, which is a significant challenge for ensuring the security of IndAI while unlocking its full potential.

3.3.5. Collaborative robots and reconfigurable executors

The design of physical infrastructures such as robots is increasingly coupled with the IndAI models. In the manufacturing paradigm evolution from mass production to mass individualization, reconfigurable and flexible manufacturing technologies such as reconfigurable collaborative robots started to emerge in the industry. An industrial robot with sensors and controllers can autonomously complete various individualized manufacturing tasks (e.g., welding, assembly, and picking) with high durability, high speed, and high precision while coping with the uncertainty of human interaction and environments [114]. In man-machine cooperative operation, the motion control of a robot needs to be limited and centered on human safety when maximizing productivity [115]. Also, incorporating IndAI into the motion control of a robot should ensure the boundedness of the system's state variables and thus enhance the stability of the system [37].

Industrial robots usually need to rely on fixed external environments. However, this kind of deployment needs a lot of material and financial resources. When the production line changes, the industrial robot also needs to be redeployed, which is costly. Collaborative robots, which can interact directly with people in collaborative areas, have the ability of rapid deployment and low comprehensive cost than traditional industrial robots. Human-robot collaboration (HRC) is identified as a key element of Industry 5.0. Adaptive control methods can enhance the flexibility of robots to humans and the environment, and reduce task-related preprogramming [116]. In HRC, the efficiency of the interaction between humans and robots will affect the overall work efficiency. Intelligent human-robot task allocation methods are supposed to allocate the sequential tasks of robots and operators in different workspaces to reduce the waste of resources [117]. Digital twins could be established for human-robot cooperative systems to enhance operation flexibility [118]. It is challenging to simultaneously balance the safety, reliability, flexibility, and other performance of the HRC.

4. Challenges and research directions for unlocking the power of IndAI

As Industry 5.0 is still in the theoretical concept stage, there will be a long time for the coexistence and interplay of Industry 4.0 and Industry 5.0. As shown in Fig. 6, there are many social barriers and technology challenges to properly selecting, designing, deploying, using, and updating various IndAI models/algorithms in practice for unlocking the power of IndAI. This section ends with a gathering of future research directions for unlocking the power of IndAI toward Industry 5.0.

4.1. Social barrier

Currently, many manufacturers have not yet fully implemented Industry 4.0 technologies, especially in small and medium-sized enterprises (SMEs) or across the value chain. To promote the Industry 5.0 vision, the reorganization of IndAI technologies must be widely implemented across the industry or in SMEs. Broader integration of associations and representatives from the industry is needed to further refine the IndAI model design, selection, and implementation.

IndAI for Industry 5.0 emphasizes the characteristics of human-centricity, sustainability, and resilience. This means that IndAI will work around workers, take humans as the center, and give full play to a human subjective initiative to improve industrial production capacity and enhance value and interests. The needs of workers of different occupations, identities, and ages are inconsistent, and the values they pursue and the opinions they consider are also very different. The value brought by new technologies cannot directly radiate to each part of society. Reaching an agreement between society and the New Deal requires broad support from society. The concept of human-centricity is thus extended to society-centricity, and the needs of individuals are supplemented by the needs of society. The individual needs of employees must be coordinated with the needs of employers and the entire workforce of society. However, for many enterprises, economic performance is the main goal of technology implementation. How to synchronize the values of enterprises and social values is a challenge. It calls for the recommended practice, policy, standard, and reference architecture to actively promote proper IndAI applications to industries, which are supposed to guide on the following issues: 1) Who using IndAI, 2) Why using IndAI, 3) Where to use IndAI, 4) What to establish IndAI, 5) When to implement IndAI, and 6) How to design and operate IndAI.

In addition, implementing IndAI may suffer from two aspects of ethics issues. The first ethical issue is that IndAI algorithms are often encapsulated as black boxes in use. Defects or incorrect values of the algorithm may trigger biased, harmful, and misleading traps [119]. For example, recommendation IndAI algorithms may tend to advertise the equipment of a certain manufacturer, resulting in a monopoly. The algorithm of human-machine collaboration may become a way for capitalists to manipulate the value of workers. AIGC algorithms may generate content that contains inaccurate operational guidelines, leading to the occurrence of hazardous accidents. The second ethical issue is the privacy risk [119]. Besides the privacy leaking caused by illegal data attacks, with the further development of intelligent sensing technologies, workers' emotions may be monitored, and deep-seated biological data may be captured, which will infringe on workers' privacy. In addition, while blockchain brings immutability of data, it also brings unforgettability of data. Human past experiences and machine operation records can be accessed, analyzed, and monitored at any time, leading to a lack of social boundaries in the human world and the leakage of core processes in manufacturing.

Possible pathways to deal with the mentioned ethical challenges include three aspects. The first one is to identify the right fundamental

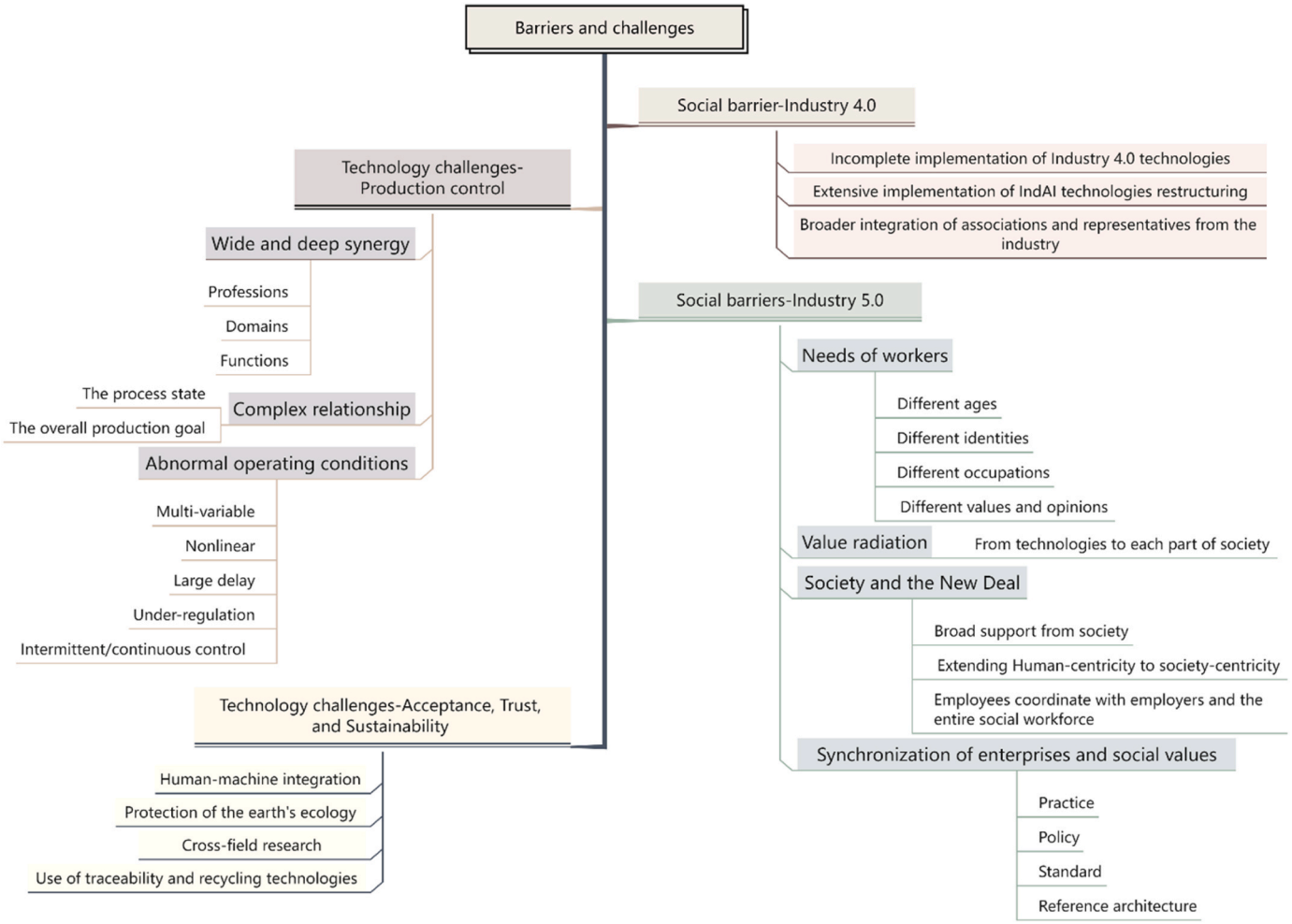


Fig. 6. Barriers and challenges in IndAI.

framework, principles, and guidelines to ensure the lifecycle ethical management (e.g., design, regulation, and use) of IndAI to benefit individuals, societies, and the environment [120]. Establishing new models to ascribe moral responsibility distributed among designers and users of IndAI is essential, and ethical analyses must be extended to account for the invisible influence exercised by IndAI on human behavior [121]. The second one is to advance technologies such as explainable IndAI, interpretable IndAI, robust IndAI, responsible IndAI, differential privacy, federated learning, and homomorphic encryption to enhance transparency, fairness, non-maleficence, responsibility, and accountability. The third pathway is a legal approach to regulate the research and application of IndAI through legislation, standards, and mandatory comprehensive evaluation. Also, Industry 5.0 claims to be a human-centric paradigm, and thereby we as humans should apply the right governance before, during, and post the implementation of the IndAI [121].

4.2. Technology challenges

From the production control perspective, it is challenging to achieve a wide as well as deep synergy across various professions, domains, and functions. The reason is that the critical operation information and important process parameters are difficult to obtain, resulting in incomplete process information and even failure of mechanism models. The relationship between the process state and the overall production goal of the whole process is complicated, so it is difficult to describe the essential characteristics of an unsteady state and nonlinearity. Within production control, the coexistence of multi-variable factors, nonlinearity, large delays, under-regulation, and intermittent/continuous control amplifies the complexity. Abnormal operating conditions are extremely difficult to detect and diagnose in the early stage, and the increasingly open industrial system leads to increased system security risks.

Also, acceptance and trust in technology are crucial. Under the human-centered idea, IndAI should work with people. In the case of this high degree of human-machine integration, IndAI technology must improve its interpretability and autonomous planning ability. IndAI must have a certain understanding of workers' behavior to better cooperate and avoid some errors that may cause harm.

Sustainability is a long-term development and a key issue in Industry 5.0. In the long run, society should not ignore the earth's ecology and must protect the earth's ecology while keeping industrial development. This requires cross-field research in combination between IndAI and nanotechnology or smart material technology to achieve environmental goals such as carbon dioxide neutralization. IndAI models are supposed to be able to make full use of traceability and recycling technologies to obtain product footprint labels so that energy can be fully utilized to meet the requirements of sustainable development.

4.3. Research directions

IndAI needs an integrated technological, organizational, policy-making, and security informatics solution. In industry, the simple pursuit of more advanced AI is not what we want to emphasize at the

moment. To further provide insights for designing appropriate models and algorithms to unlock the power of IndAI in the process of digital transformation and upgrading of the industry, future directions are gathered based on value-oriented awareness as well as a systematic understanding of IndAI technologies (Fig. 7).

4.3.1. Rapid response and reliability

Along with the operation of IndAI algorithms, the surge of data has brought a huge burden to servers and networks. It faces the risk of delay and latency [122]. To alleviate this situation, edge intelligence [123] has been gradually studied and applied. The intelligence at the edge of industrial systems is improved to optimize resource allocation and solve problems characterized by diversity and time variability. Three directions to improve the rapidity of IndAI systems could be concluded as follows: 1) How to synthesize cognitive information/matter/energy flows and their interaction mechanisms in real time? 2) How to model the multi-scale correlation and establish the multi-spatial and temporal panoramic features of industrial systems? and 3) How to solve the cooperative regulation under complex constraints to achieve efficient comprehensive optimization?

In addition, edge devices are also vulnerable to malicious nodes, and it is a great challenge to ensure the security of edge services. Through the large-scale, ubiquitous, and networked deployment of detection and control systems, a sharing and coordination mechanism that integrates across time, space, system levels, and borders could be formed. A highly permeable, flexible, reconfigurable, and community-based perception and regulation network could be established. Integrating edge intelligence with blockchain authorization could achieve flexible and secure edge service management [124]. Distributed deep learning could also be established for attack detection on edge devices to improve system stability and reliability [125].

4.3.2. Security in a distributed computing context

Industrial data usually exists in the form of distributed silos, and direct data exchange between silos is difficult due to security and privacy issues. The emergence of secure multi-party computing, verifiable computing, and federated learning [126] provides a secure solution to this problem. Federated learning-based AI (here it is defined as federated intelligence) can train IndAI models based on distributed encrypted datasets without having to centralize all data to a central storage node. Federated intelligence allows knowledge sharing without compromising user privacy. However, federal intelligence cannot fully guarantee the security of data [127]. By sharing encrypted data, adversaries can still attack it, destroying its security and causing information leakage [128]. Blockchained privacy-preserving data-sharing mechanisms could enhance the security of federated intelligence. Security enhancement that deeply integrates functional security and information security should be performed to form a comprehensive predictable and highly reliable technical system of risk assessment, vulnerability analysis, vulnerability mining, countermeasure protection, health diagnosis, and monitoring of industrial systems.

4.3.3. System adaptability

Adaptability is a kind of ability to cross professional boundaries and be competent for critical applications. It can deeply integrate complex mechanism characteristics and dynamic changing environment to deal with strong coupling, nonlinear, large time delay, and harsh constraints, and carry out real-time dynamic decision-making while ensuring the system stability, controllability, optimality, and robustness. The adaptability of IndAI models is reflected in three aspects, namely, model interaction, hardware/software adaptation, and data/task adaptation of algorithms.

In terms of interaction between models, a variety of learning frameworks/paradigms were designed to implement practical IndAI applications, and the underlying implementation technologies of each framework are different, resulting in difficulties in mutual conversion between IndAI models. A direct model transformation is usually more complex than the development of a general framework, involving the inherent characteristics of the industrial field [129].

The second is software/hardware adaptation. The development of CPU and software frameworks is different, which leads to the mismatch between the underlying hardware infrastructure and the software framework, making IndAI algorithms often encounter some inexplicable errors. Therefore, software framework developers and CPU manufacturers need to ensure good software/hardware compatibility. A compiler for IndAI algorithms/models is needed to solve the adaptation and portability between different software frameworks and different hardware devices. However, there is no uniform standard for such compilers, and the competition among hardware manufacturers in the middle presentation layer has also become an obstacle to the practical application of IndAI technologies.

The third is data/task adaptation. At present, IndAI algorithms usually rely on a large number of labeled data for training. With the gradual acceleration of the penetration of AI and industrial integration, more and more application scenarios are facing the problem of “small data”. In addition, due to the differentiation of industrial equipment products, scenarios, and the diversity of tasks, the replicability of IndAI models in current industrial scenarios is poor, such as fault diagnosis in

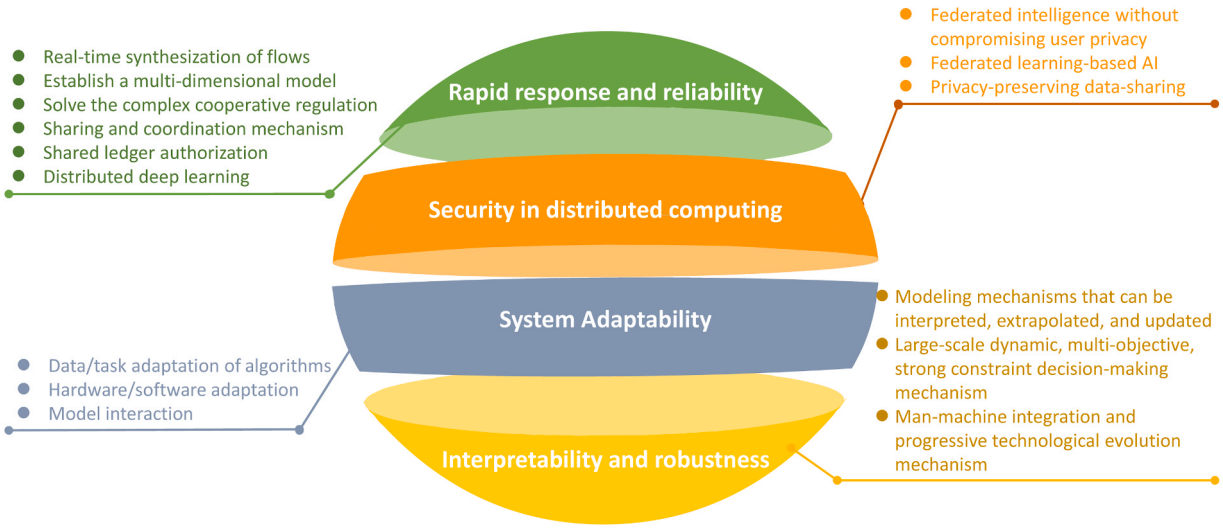


Fig. 7. Research directions of IndAI.

industrial scenarios. The types of defects in different production lines and different products may be completely different. Efficient algorithms or solutions may be obtained in one enterprise, but they perform poorly when being transferred to another enterprise.

The adaptability may also originate from the active integration of IndAI models with industrial processes, equipment, material, and structural design in the design and development of the IndAI system. In response to changes in tasks, environment, systems' characteristics, and even value system changes, adaptability is enhanced by a constantly continuous self-evolving and enhancement mechanism via deep processing of information, real-time dynamic feedback, system optimization decision-making, and forward-looking global response.

4.3.4. *Interpretability and robustness*

Many existing IndAI models are "black box" that lacks interpretability [130]. At present, although neural networks have shown good results on complex problems, "connectionism" represented by neural networks cannot provide a clear semantic explanation [131]. Almost all deep learning breakthroughs are essentially curve-fitting [132]. Since operators are not clear about their operating mechanism and the reasons for making wrong decisions, there may be problems in adopting this model in key and sensitive areas [133]. The solution must be reliable and robust, that is, the mechanism behind it needs to be clearly explained [134]. It is of great significance to systematically summarize and visualize the features learned by IndAI models and how these features interact to predict [135]. It may be concluded into three aspects to achieve this goal: 1) Modeling mechanisms that can be interpreted, extrapolated, and updated; 2) Large-scale dynamic, multi-objective, strong constraint decision-making mechanism; and 3) Man-machine integration and progressive technological evolution mechanism.

5. Concluding remarks

This perspective article presents a classification of IndAI based on the intelligence levels (i.e., Sharable IndAI, Autonomous IndAI, Learnable IndAI, Knowledgeable IndAI, and Reactive IndAI) and gives four principles for implementing IndAI, which provides the guide for designing and implementing appropriate models and algorithms in practice. Three significant opportunities of IndAI, namely, collaborative intelligence, self-learning intelligence, and crowd intelligence, towards Industry 5.0 vision are identified to promote the transition from a technology-driven initiative in Industry 4.0 to the coexistence and interplay of Industry 4.0 and a value-oriented proposition in Industry 5.0. Pathways of IndAI applications in Industry 4.0 are discussed in five dimensions, namely, design, manufacturing, marketing and logistics, operation and maintenance service, and recycling and remanufacturing. Future research directions of IndAI are identified as four aspects, rapid response and reliability, security in a distributed computing context, adaptability, and interpretability and robustness. This perspective article is a reflection of significant conceptual and technical advances in unlocking the power of IndAI in this junction of dramatic development of Industry 4.0 and Industry 5.0. We hope that our effort lays a strong foundation for making IndAI research and engineering more sustainable, resilient, and human-centric.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

This work was supported by the National Key Research and Development Program of China under Grant No. 2022YFF0606005; the National Natural Science Foundation of China under Grant No. 52075107 and U20A6004; and the Natural Science Fund of Guangdong Province under Grant No. 2022B1515020006.

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